

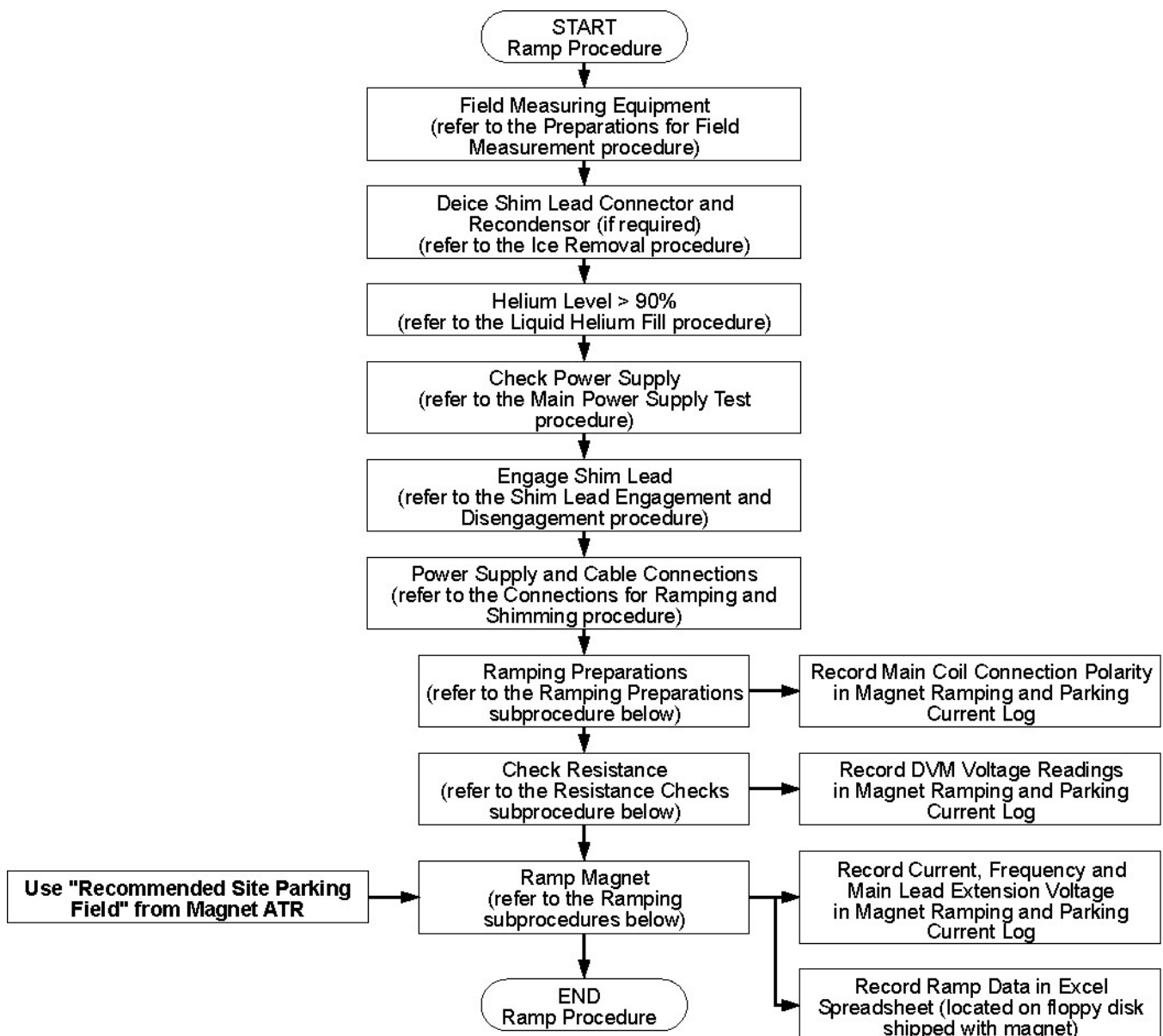
7 Ramping Magnet to Field

7.1 Ramping Introduction

LCC magnets are ramped using voltage control to a nominal current of 736 amps for 1.0T LCC's or 740 amps for 1.5T LCC's.

A 750 Amp Ramp Cart/Cable Kit (2135435) or a 1000 Amp Ramp Cart/Cable Kit w/ Ground (2353394) plus a Field Shim Case & Cable Kit (2135558) are required to ramp and shim the magnet. Additionally, a MetroLab Teslometer Kit (46-251865G4) is required, although Universal Field Mapping Fixture 46-294060G4 is not required since only the overall field strength is monitored during ramping.

Illustration 3-75: Ramp Flowchart



List of Procedures & Data Sheets Referenced
Section 4, Preparations for Field Measurement
Chapter 5, Section 9.2, Ice Removal
Section 3.2, Liquid Helium Fill
Chapter 4, Section 8.1, Main Power Supply Test
Section 5, Shim Lead Engagement and Disengagement
Section 6, Connections for Ramping and Shimming
Section 7.4.1, Ramping Preparations
Section 7.3, Resistance Checks
See Section 7.4.1, List of Ramping Procedures
Chapter 8, Section 5, Magnet Ramping and Parking Current Log

7.2 Ramping Preparations

7.2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.2.2 Overview

The "Personnel Requirements" and "Preliminary Requirements" stated here are for the entire ramping procedure.

7.2.3 Preliminary Requirements

7.2.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Nonabsorbent protective clothing (long sleeve shirt and pants)	1 set per person	-	-	-
Nonferrous Shoes	1 pair per person	-	-	-
750 Amp TCR 7.5T750 Main Power Supply (46-260776G3) or 1000 Amp ESS 7.5-1000-2-D-1236 Main Power Supply (46-260776G4)	1	-	-	-
750 Amp Ramp Cable Kit (2135435) or 1000 Amp Ramp Cable Kit w/ Ground (2353394)	1	-	-	-
Fixed Main Lead Extension (46-294204G1) or Flexible Main Lead Extension (2287029) (for low ceiling height sites)	2	-	-	-
Hold-Down Tool (2193922) or Hold-Down Tool, Short (2276755) (for low ceiling height sites)	1	-	-	-

Item	Qty	Effectivity	Part#	Manufacturer
Retractable Shim Lead Assembly (2173354) or Retractable Shim Lead Assembly, Short (2285073) (for low ceiling height sites)	1	-	-	-
Digital Voltmeter (DVM)	1	-	-	-
Warning Sign & Label Kit	1	-	46-258770G4	-
Portable Oxygen Monitor	1	-	2287000	-
Cryogen Safety Wear Kit	1 kit per person	-	46-271137G1	-
Field Shim Cable Kit (if not already ordered with the magnet)	1	-	2135558	-
MetroLab Teslameter Kit	1	-	46-251865G4	-
#5 MetroLab Probe (included in MetroLab Teslameter Kit 46-251865G4)	1	-	46-255839P104	-
Nonmagnetic Tools Kit	AR	-	46-320273G4	-

7.2.3.2 Consumables

No consumables required.

7.2.3.3 Replacement Parts

No replacement parts required.

7.2.3.4 Safety



⚠ DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, BEFORE STARTING THIS PROCEDURE.



⚠ DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS, BEFORE AND WHILE PERFORMING THIS PROCEDURE.



! DANGER

POTENTIAL FATAL INJURY!

HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION, BEFORE STARTING THIS PROCEDURE.



! WARNING

POTENTIAL PROJECTILE HAZARD

FERROMAGNETIC OBJECTS IN A STRONG MAGNETIC FIELD CAN BECOME DANGEROUS PROJECTILES.

REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. MAKE SURE ALL EQUIPMENT AND TOOLS USED IN THE MAGNET ROOM ARE NON-FERROMAGNETIC. LOCATE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND ROOM GEOMETRY ALLOWS. DO NOT ALLOW ANYONE CARRYING FERROMAGNETIC MATERIALS — INCLUDING CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES — INSIDE THE 5 GAUSS ZONE.



! WARNING

HIGH MAGNETIC FIELD

CARDIAC PACEMAKERS, NEUROSTIMULATORS (OR OTHER BIOSTIMULATORS), STEEL PLATES AND FERROMAGNETIC MATERIAL, EQUIPMENT AND TOOLS WILL BE ADVERSELY EFFECTED BY THE STRONG MAGNETIC FIELD.

□□□ NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN. □□□ POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR OTHER BIOSTIMULATORS NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS. □□□ POST “RAMPED MAGNET” AND “AUTHORIZED PERSONNEL ONLY” WARNING SIGNS AT THE MAGNET ROOM ENTRANCE AND AT THE MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.



! WARNING

ASPHYXIATION HAZARD!

THIS PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR.

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON — OR THE HATCH IS OPENED IF A MOBILE VAN — BEFORE STARTING THIS PROCEDURE.


! WARNING

EXPLOSION HAZARD!
LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN, PRODUCING A HIGHLY COMBUSTIBLE OXYGEN LIQUID.
NO SMOKING IN MAGNET AND CRYOGEN STORAGE ROOMS!


! WARNING

HIGH MAGNETIC FIELD
BE PREPARED FOR EMERGENCIES
MAKE SURE THE MAGNET RUNDOWN UNIT (MRU) IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE "MAGNET RUNDOWN UNIT (MRU) INSTALLATION" IN THE "MAGNET SYSTEM INSTALLATION" SECTION EARLIER IN THIS CHAPTER.


! WARNING

POTENTIAL MAGNET QUENCH!
THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES, IF ITS LIQUID HELIUM LEVEL DROPS TOO LOW DURING THIS PROCEDURE.
MAKE SURE THE MAGNET $\geq 90\%$ FULL OF LIQUID HELIUM BEFORE STARTING THIS PROCEDURE.


! WARNING

POTENTIAL COLD BURN HAZARD
EXPOSURE TO CRYOGENS AND CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE.
WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHEN PERFORMING THIS PROCEDURE.

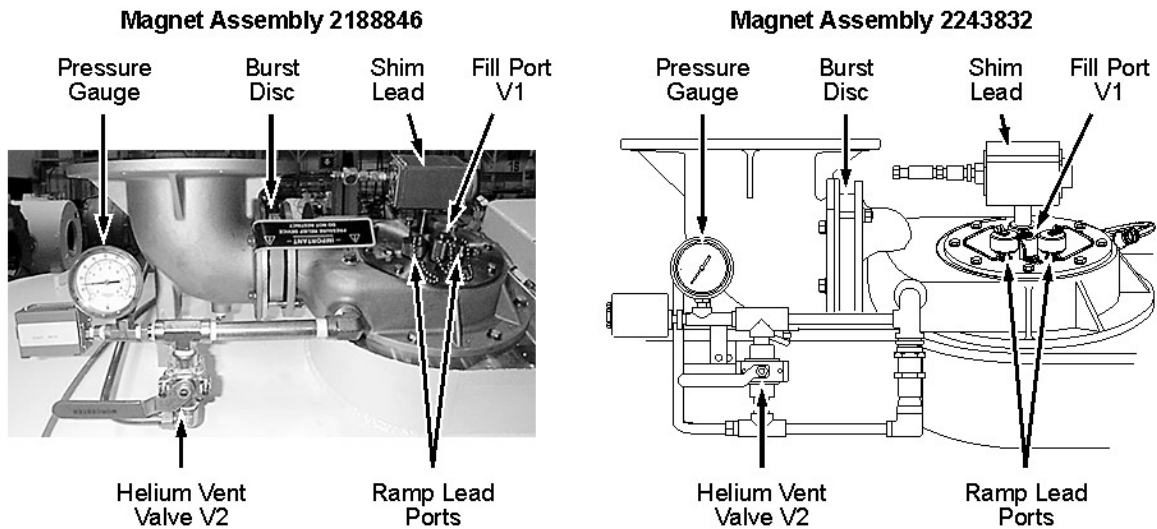
7.2.3.5 Required Conditions

Condition	Reference	Effectivity
Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.	-	-
The removal of some magnet enclosure components may be required to complete this procedure. Refer to Chapter 5, Section 5, Upper and Side Enclosure Removal, before continuing with this procedure if cover removal is required.	-	-

Condition	Reference	Effectivity
The Shim Lead Connector must be free of ice. Perform Shim Lead Connector checks in conformance with Chapter 4, Section 5, Magnet Electrical Checks. If an ice blockage is found, deice in conformance with the Shim Lead Connector and Vertical Penetration Deicing subsection of Chapter 5, Section 9.2, Ice Removal.	-	-
Purge the Recondenser with helium gas in conformance with the Level Three Deicing subsection of Chapter 5, Section 9.2, Ice Removal.	-	-

7.2.4 Procedure

1. Make sure helium level readings on P403-1 and P403-2 are functional and accurate. If problems are found, troubleshoot before ramping.
2. Set up the MetroLab Teslameter (46-255839P2) and a #5 Teslameter Probe (46-255839P104), for general field strength monitoring in conformance with Section 4, Preparations for Field Measurement.
3. Check for ice on the Shim Lead Connector and the Recondensor. Deice in conformance with the Shim Lead Connector and Vertical Penetration Deicing subsection of Chapter 5, Section 9.2, Ice Removal, before ramping the magnet.
4. Make sure the magnet is > 90% full of helium and Vent Valve V2 (Illustration 3-76) is closed before ramping the magnet. If necessary, fill the magnet in conformance with Section 3.2, Liquid Helium Fill.

Illustration 3-76: Helium Vent Plumbing**CAUTION****Potential Magnet Quench**

The magnet may quench if the Main Power Supply experiences large output fluctuations and/or excessive ripple.

Make sure the Main Power Supply is routinely calibrated at an approved facility.

5. Make sure that the Main Power Supply is installed, checked and adjusted in conformance with the vendor manual supplied with the unit and either Chapter 4, Section 8.2, TCR 7.5T750 Main Power Supply Tests, for 750 amp power supplies or Chapter 4, Section 8.3, ESS7.5-1000-2-D-1236 Main Power Supply Tests, for 1000 amp power supplies, as appropriate.
6. Make sure the Shim Lead Assembly is "Engaged" in conformance with Section 5, Shim Lead Engagement and Disengagement.
7. Make sure the Input Power Cable to the Main Power Supply is disconnected.

**WARNING****POTENTIAL MAGNET QUENCH!**

IMPROPER ENGAGEMENT/DISENGAGEMENT OF THE MAIN LEAD EXTENSIONS WITH THE MAIN LEAD PINS CAN CAUSE A RAMPED MAGNET TO QUENCH. INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW THEM TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE THEY CONTACT THE MAIN LEAD PINS.

**! WARNING**

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING**

ELECTRICAL SHOCK HAZARD!
DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**! WARNING**

POTENTIAL COLD BURN HAZARD
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

8. Connect the power supply to the magnet by making all cable connections in conformance with Section 6, Connections for Ramping and Shimming.
9. Record the Main Coil connection polarity in Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
10. Set all heater switches on the power supply (Illustration 3-77: 750 amp; Illustration 3-78: 1000 amp) to the "OFF" position. Set the Current Adjust and Voltage controls to zero (full CCW).

Illustration 3-77: 750 Amp TCR 7.5T750 Superconducting Main Coil Service Power Supply Cabinet (46-260776G3)

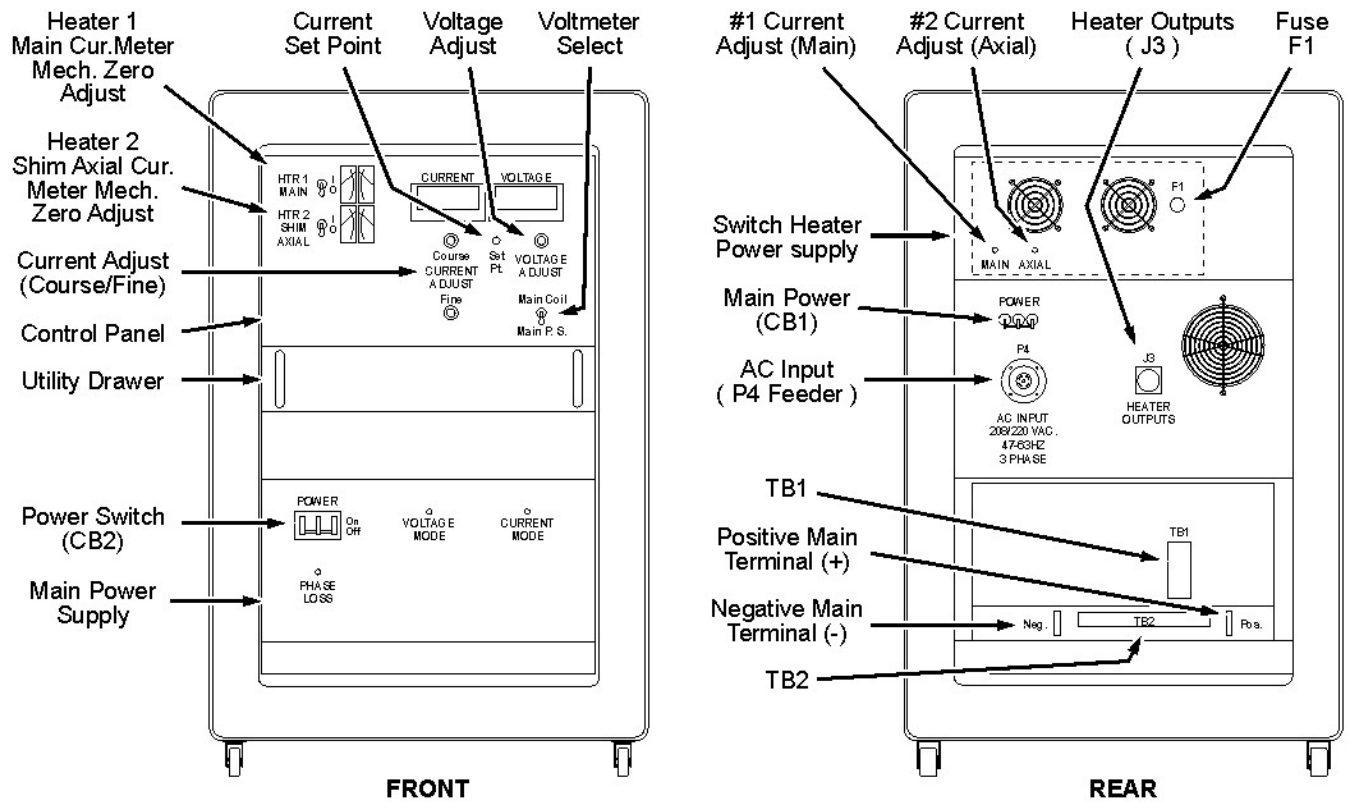
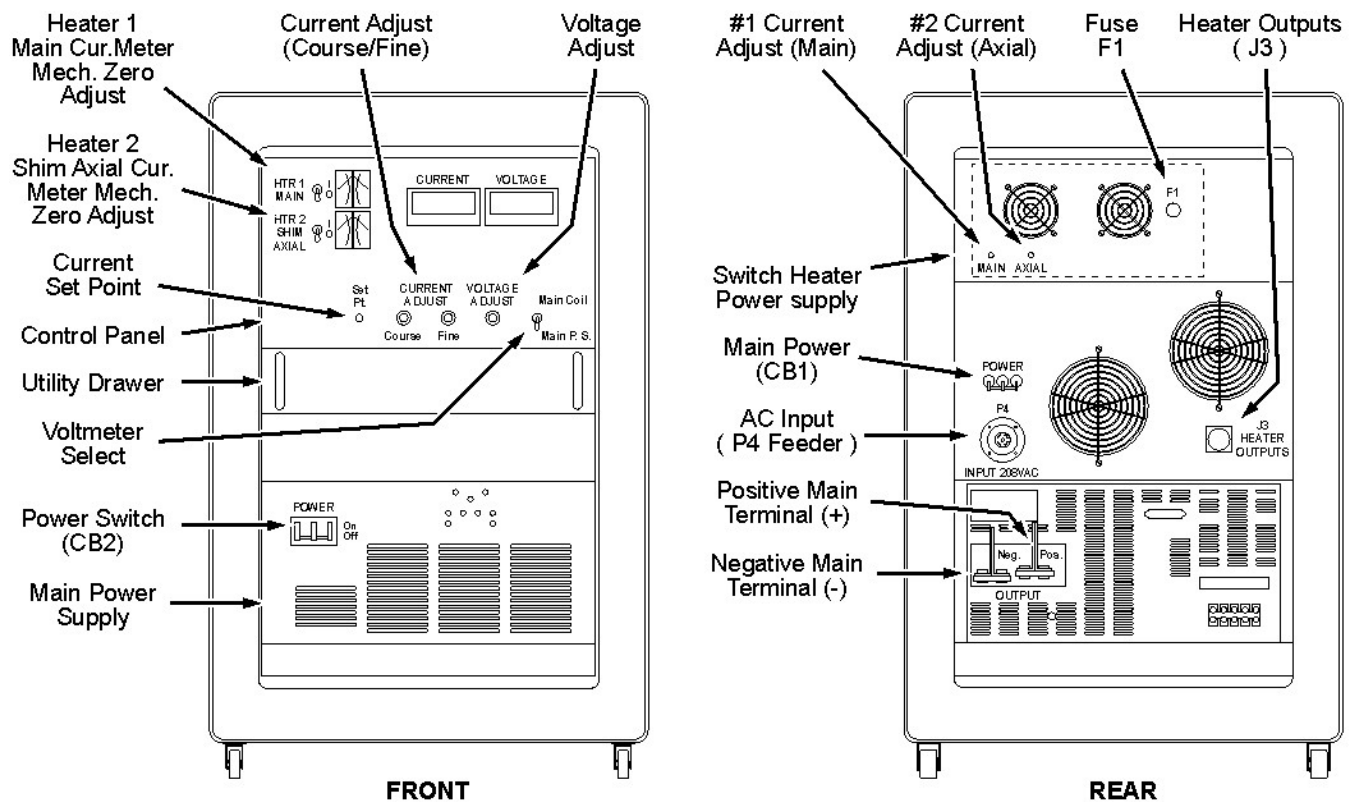


Illustration 3-78: ESS7.5-1000-2-D-1236 Superconducting Main Coil Service Power Supply Cabinet (46-260776G4)



11. Connect the Input Power Cable to the Main Power Supply.

7.2.5 Finalization

No finalization steps.

7.3 Resistance Checks

7.3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.3.2 Overview

See Section 7.2, Ramping Preparations, for "Personnel Requirements" and "Preliminary Requirements" for the entire ramping procedure.

7.3.3 Preliminary Requirements

7.3.3.1 Tools and Test Equipment

No tools or test equipment required.

7.3.3.2 Consumables

No consumables required.

7.3.3.3 Replacement Parts

No replacement parts required.

7.3.3.4 Safety

No general safety information.

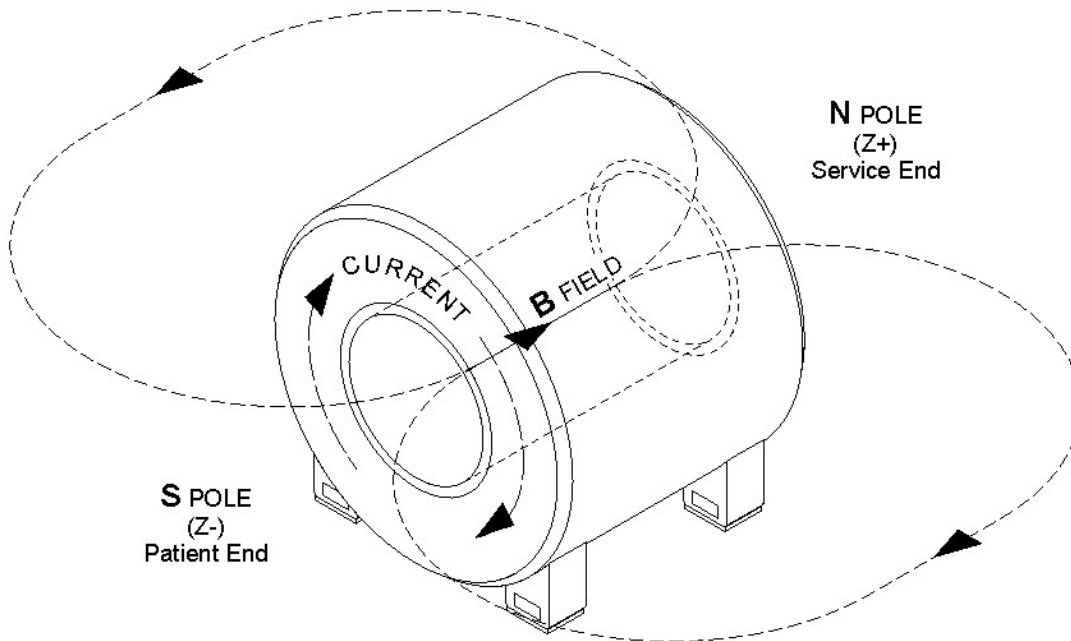
7.3.3.5 Required Conditions

No required conditions.

7.3.4 Procedure

1. If magnet is not at field, check Switch Heater and Shim Coil resistances in conformance with Chapter 4, Section 5, Magnet Electrical Checks.
2. Make sure the Current Adjust and Voltage Adjust controls on the Main Power Supply are "OFF" (full CCW).

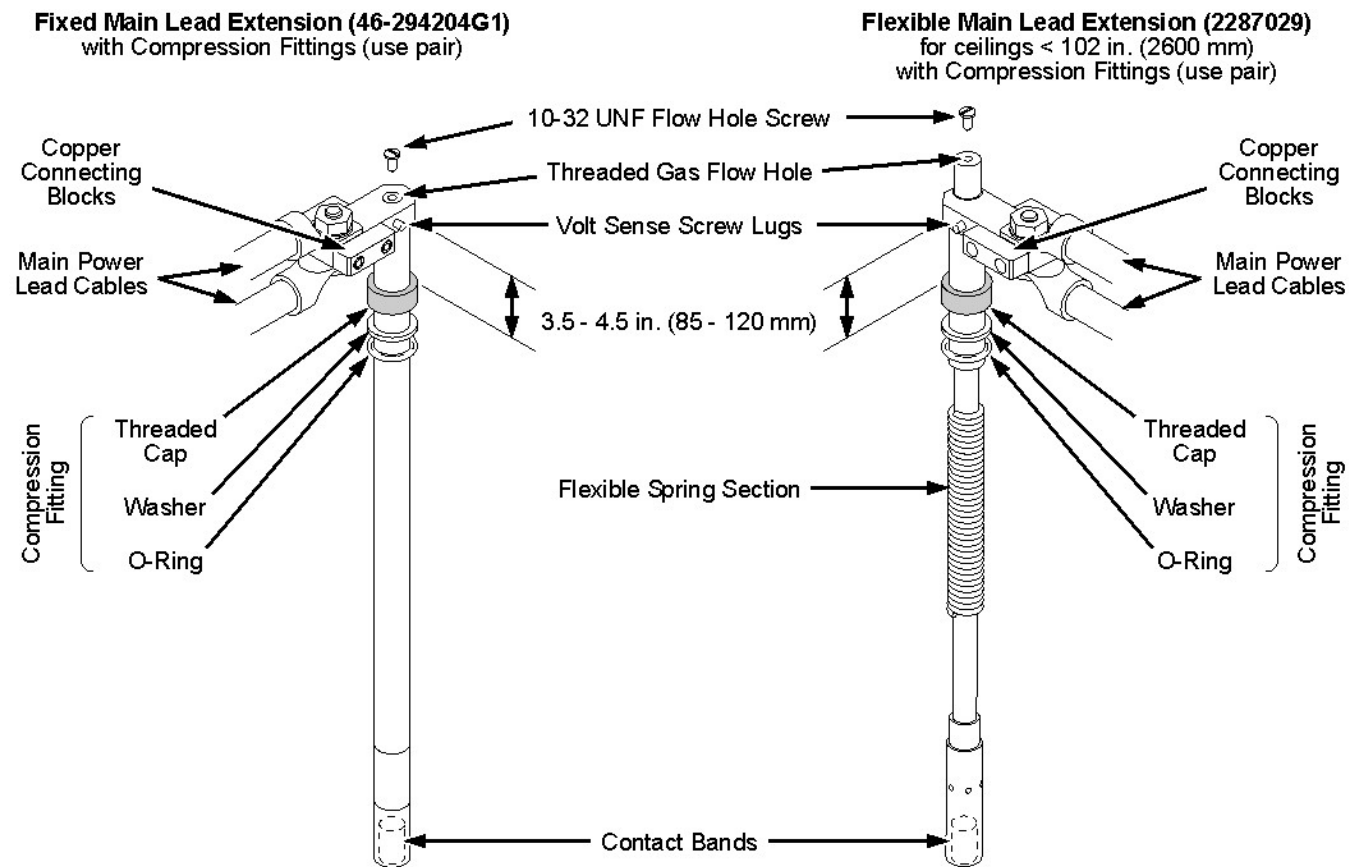
NOTE: All main coil driving voltages provided in this procedure will be equal in magnitude but opposite in polarity for "reverse ramped" magnets. Normal polarity is shown in Illustration 3-79.

Illustration 3-79: Magnet Polarity Ramped Normal

3. Set the Main Power and Power On switches to "ON."

NOTE: There are duplicate Main Power and Power On switches on the front and back of the Main Power Supply.

4. Make sure the Main Heater Switch is off, then set the Heater 2 Shim Axial Switch to "I" (on) and observe the current rise in the Main Power Supply Ammeter (800 - 820 mA) to verify circuit continuity.
5. Connect a Digital Voltmeter (DVM) to the end of the Voltage Sense Leads.

Illustration 3-80: Main Lead Extensions

6. Set the Current Adjust Coarse Control on the power supply to maximum (full CW). See Illustration 3-81 and Illustration 3-82.

Illustration 3-81: 750 Amp TCR 7.5T750 Superconducting Main Coil Service Power Supply Cabinet (46-260776G3)

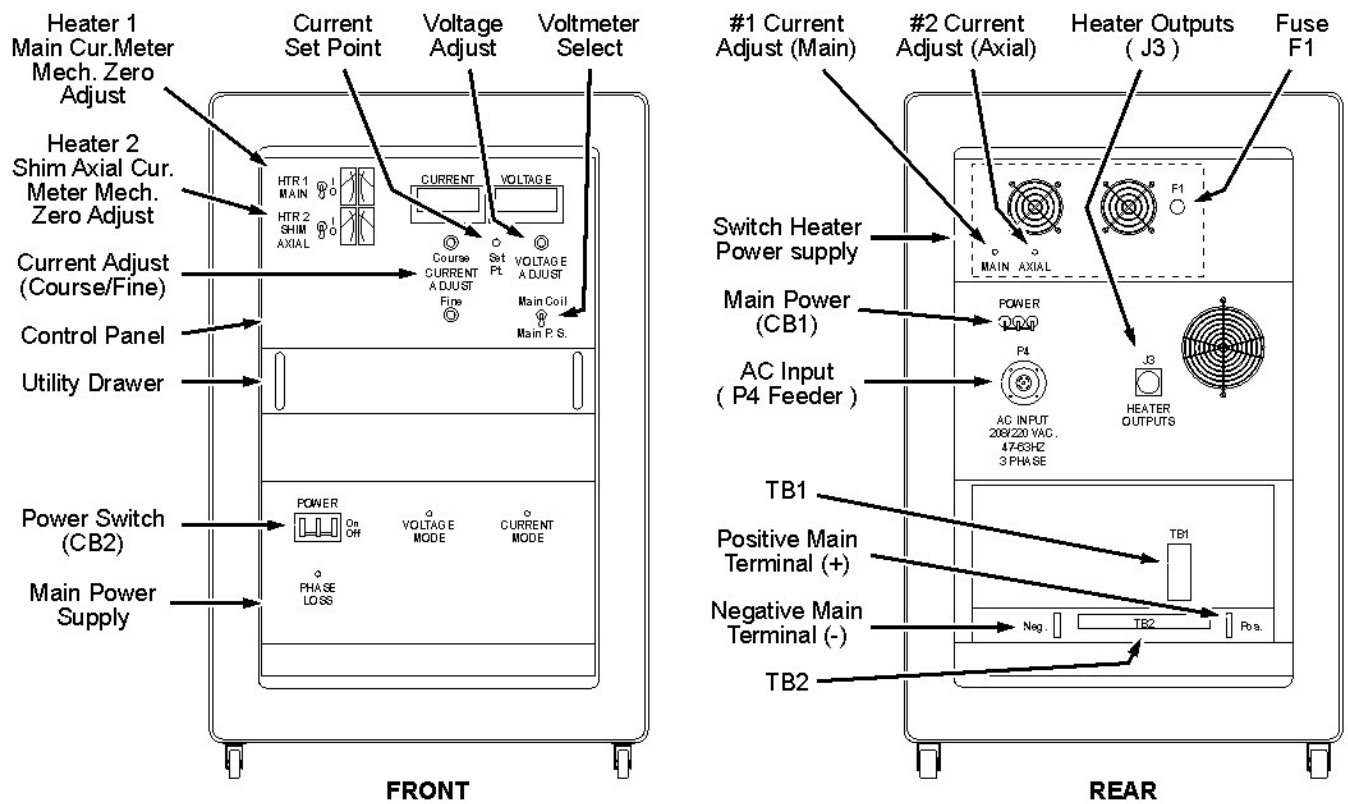
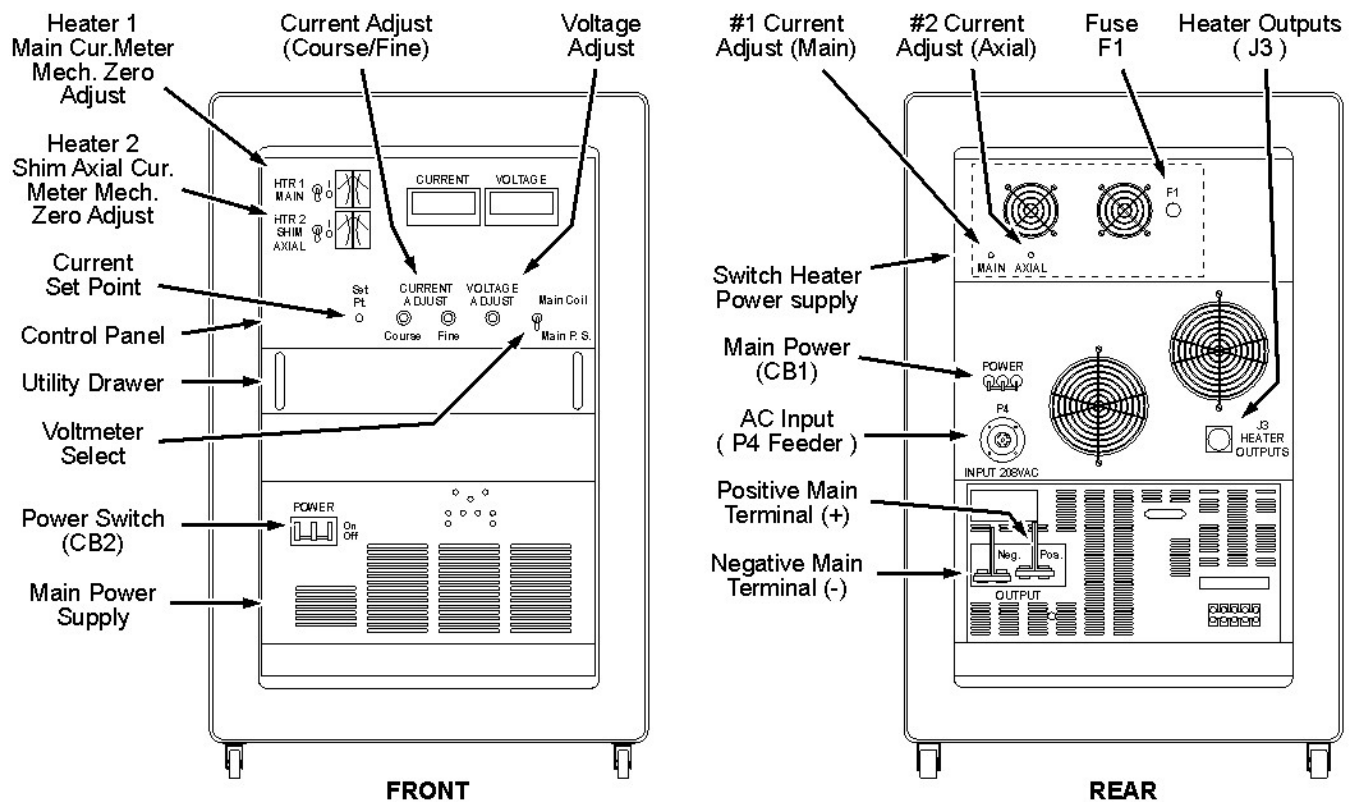


Illustration 3-82: ESS7.5-1000-2-D-1236 Superconducting Main Coil Service Power Supply Cabinet (46-260776G4)



7. Observe the Main Power Supply Ammeter and slowly turn the Voltage Adjust Control clockwise to set a 750 amp current through the Main Leads, Main Lead Extensions and persistent Main Switch.
8. Record the voltage reading on the DVM in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.



CAUTION

Potential Magnet Quench

Voltage readings > 150 millivolts at 750 amps indicate unacceptable internal contact resistance at the Main Lead Extensions. Higher resistances will add more heat to the magnet, increasing boiloff and possibly causing a quench during ramping. Make sure the voltage readings are ≤ 150 mV at 750 amps.

9. If the DVM voltage is > 150 millivolts, perform one or more of the following steps to reduce contact resistance:
 - Wait approximately 1 minute with current running, readings may drop as the Main Lead Extensions cool.

- Tighten the nuts (one flat at a time) on top of the Hold-Down Tool while measuring DVM readings.

If the reading still exceeds 150 mV:

1. Turn the Voltage and Current Adjust controls to zero (full CCW).
2. Turn off the Magnet Power Supply input power.
3. Tighten the bolts securing the Ramp Cables to the Power Supply and Main Leads Extensions.
4. Lift and reseal the Main Leads.

Repeatedly failing the contact resistance check may indicate one of the following:

- The Main Lead Extension Contact Bands replaced as directed Section 6, Connections for Ramping and Shimming, in are not performing correctly and need replacing in conformance with Chapter 5, Section 4, Main Lead Extension Contact Band Replacement.
- The Main Leads are damaged and need replacing.

If changes are made in Step 9, repeat Step 6 through Step 9.

10. Continue to Step 11 after the DVM voltage is < 150 mV.
11. Set the power supply's Voltmeter Select Switch to the "MAIN POWER SUPPLY" position. This will display the output at the output lugs of the power supply monitored. A voltage < 2.2 V at 750 A indicates acceptable system resistance. If the voltage exceeds 2.2 V during the test, follow the procedures in Step 9 for decreasing contact resistance.
12. Gradually increase the Voltage Adjust Control to pass 750 A through the Main Power Leads, Lead Extensions and persistent Main Switch while observing the power supply's Voltmeter. If the voltage exceeds 2.2 V, then retighten the bolts securing the Ramp Cables to the Power Supply and Main Lead Extensions.
13. Turn the Current Adjust and Voltage controls off (full CCW).

7.3.5 Finalization

No finalization steps.

7.4 Ramping

7.4.1 List of Ramping Procedures



WARNING

POTENTIAL QUENCH HAZARD
RAMPING A LCC MAGNET USING AN INCORRECT RAMPING PROFILE MAY
CAUSE A MAGNET QUENCH.
MAKE SURE THE CORRECT RAMPING PROFILE FOR THE MAGNET MODEL AND
SITE APPLICATION FROM THE TABLE BELOW IS FOLLOWED.

Table 3-10: Correct Ramping Sections for LCC Magnets

Magnet Model	Magnet Version (Part Number)	Ramp In Conformance With
1.0T LCC	Fixed Site (2226333)	Section 7.4.2, 1.0T Fixed Site Ramping
	Mobile Site (2226334)	Section 7.4.5, 1.0T Mobile Site Ramping
	QR Series Magnet (2243334)	Section 7.4.4, 1.0T QR Series Ramping
	YQ Series Magnet (Fixed Site: 2226333-2; Mobile Site: 2226334-2)	Section 7.4.7, 1.0T YQ Series Ramping
1.5T LCC	Fixed Site (2226300)	Section 7.4.3, 1.5T Fixed Site Ramping
	Mobile Site (2226322)	Section 7.4.6, 1.5T Mobile Site Ramping
	YR Series Magnet (Fixed Site: 2226300-2; Mobile Site: 2226322-2)	Section 7.4.8, 1.5T YR Series Ramping

7.4.2 1.0T Fixed Site Ramping

7.4.2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.2.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.2.3 Preliminary Requirements

7.4.2.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.2.3.2 Consumables

No consumables required.

7.4.2.3.3 Replacement Parts

No replacement parts required.

7.4.2.3.4 Safety

No general safety information.

7.4.2.3.5 Required Conditions

No required conditions.

7.4.2.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:
 - **Voltage Adjust Control:** zero (full CCW)
 - **Current Adjust Control:** maximum (full CW)
 - **Main Power Switches on the Main Ramp Power Supply:** "ON"
 - **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
 - **Voltmeter Select Switch on the power supply:** "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).
4. Ramp the magnet in conformance with Table 3-11 as follows:
 - a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 5.00 volts.
 - b. Allow the voltage to decay over the current interval shown in the table.
 - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-11: 1.0T Fixed Site Ramping Profile

"Stage"	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	5.00	0 - 100	
2	4.60	100 - 200	
3	4.20	200 - 300	
4	3.80	300 - 400	
5	3.30	400 - 500	
6	2.50	500 - 550	
7	2.00	550 - 600	
8	1.50	600 - 625	
9	1.00	625 - 650	
10	0.75	650 - 675	
11	0.60	675 - 700	
12	0.50	700 - 720	
13	0.30	720 - 730	
14	Slowly lower from 0.30 Volts to 0.10 Volts	730	
15	0.10	—	0.999
16	Set voltage as required to maintain constant field	Use "Recommended Site Parking Field" frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 4.8 - 5.2 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



! WARNING

POTENTIAL MAGNET QUENCH!

1.0T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 730 amps, slowly lower the Voltage Adjust Control from 0.3 V to 0.1 V Main Coil Voltage.

8. When the field strength reaches 0.999 Tesla, slowly adjust the Voltage Adjust Control to a 0.1 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.0T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 745 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.

13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).
16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.
17. Remove the Ramp Lead Hold-Down Tool.

**⚠ WARNING**

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**⚠ WARNING**

ELECTRICAL SHOCK HAZARD!
DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

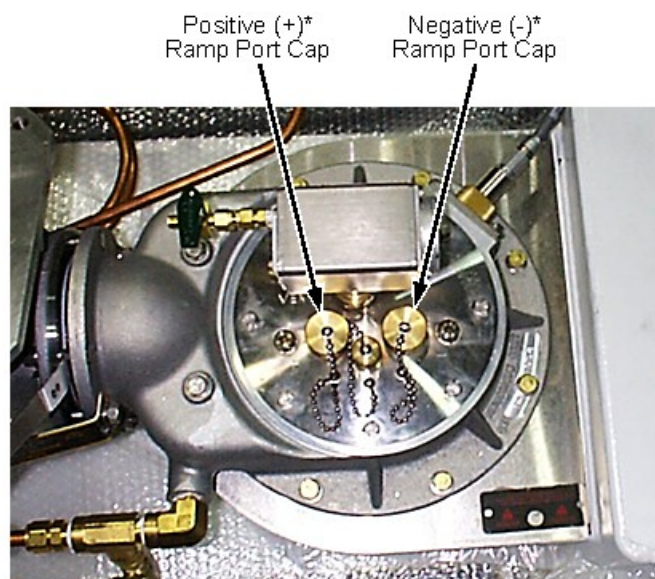
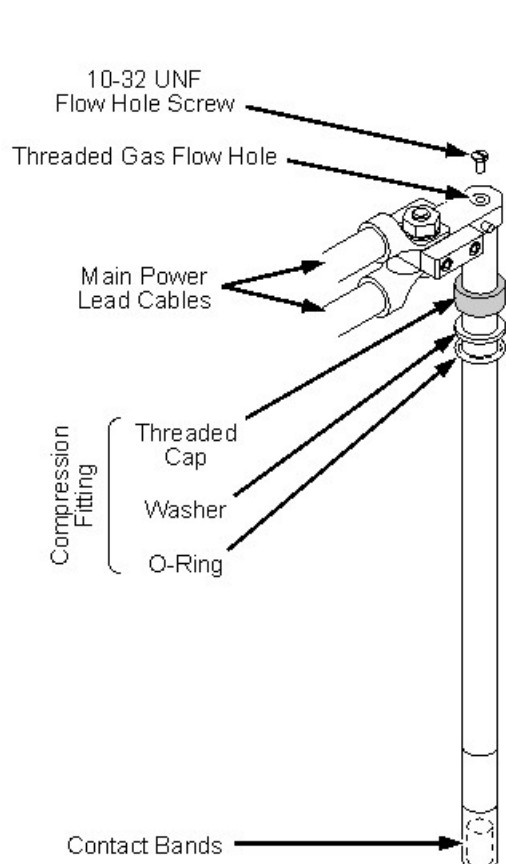
**⚠ WARNING**

POTENTIAL COLD BURN HAZARD
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:
 - a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
 - b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-83.

Illustration 3-83: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.

**NOTICE**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.

**NOTICE**

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.2.5 Finalization

No finalization steps.

7.4.3 1.5T Fixed Site Ramping

7.4.3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.3.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.3.3 Preliminary Requirements

7.4.3.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.3.3.2 Consumables

No consumables required.

7.4.3.3.3 Replacement Parts

No replacement parts required.

7.4.3.3.4 Safety

No general safety information.

7.4.3.3.5 Required Conditions

No required conditions.

7.4.3.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



⚠ CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:
 - **Voltage Adjust Control:** zero (full CCW)
 - **Current Adjust Control:** maximum (full CW)
 - **Main Power Switches on the Main Ramp Power Supply:** "ON"
 - **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
 - **Voltmeter Select Switch on the power supply:** "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).
4. Ramp the magnet in conformance with Table 3-12 as follows:
 - a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 6.00 volts.
 - b. Allow the voltage to decay over the current interval shown in the table.
 - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-12: 1.5T Fixed Site Ramping Profile

"Stage"	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	6.00	0 - 100	
2	5.60	100 - 200	
3	5.20	200 - 300	
4	4.70	300 - 400	
5	4.30	400 - 500	
6	3.50	500 - 550	
7	3.00	550 - 600	
8	2.35	600 - 625	
9	2.00	625 - 650	
10	1.00	650 - 675	
11	0.75	675 - 700	
12	0.50	700 - 720	
13	0.30	720 -730	
14	Slowly lower from 0.30 Volts to 0.10 Volts	730	
15	0.10	—	1.499
16	Set voltage as required to maintain constant field	Use "Recommended Site Parking Field" frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 10.9 - 11.3 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



! WARNING

POTENTIAL MAGNET QUENCH!

1.5T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 730 amps, slowly lower the Voltage Adjust Control from 0.3 V to 0.1 V Main Coil Voltage.

8. When the field strength reaches 1.499 Tesla, slowly adjust the Voltage Adjust Control to a 0.1 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.5T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 742 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.

13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).
16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.
17. Remove the Ramp Lead Hold-Down Tool.

**! WARNING**

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING**

ELECTRICAL SHOCK HAZARD!
DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

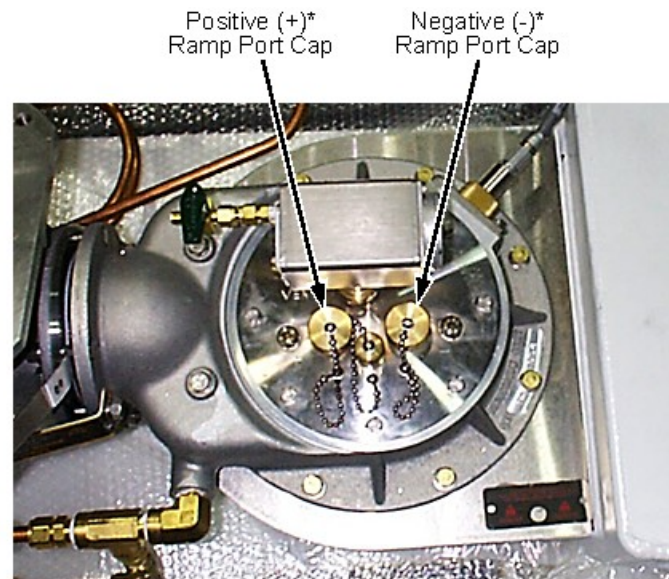
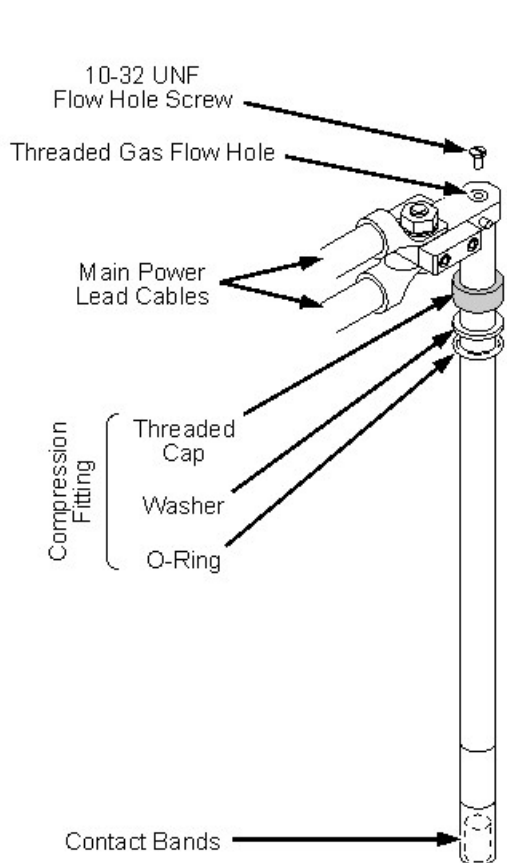
**! WARNING**

POTENTIAL COLD BURN HAZARD
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:
 - a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
 - b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-84.

Illustration 3-84: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.

**NOTICE**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.

**NOTICE**

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.3.5 Finalization

No finalization steps.

7.4.4 1.0T QR Series Ramping

7.4.4.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.4.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.4.3 Preliminary Requirements

7.4.4.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.4.3.2 Consumables

No consumables required.

7.4.4.3.3 Replacement Parts

No replacement parts required.

7.4.4.3.4 Safety

No general safety information.

7.4.4.3.5 Required Conditions

No required conditions.

7.4.4.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:
 - **Voltage Adjust Control:** zero (full CCW)
 - **Current Adjust Control:** maximum (full CW)
 - **Main Power Switches on the Main Ramp Power Supply:** "ON"
 - **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
 - **Voltmeter Select Switch on the power supply:** "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 300 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).
4. Ramp the magnet in conformance with Table 3-13 as follows:
 - a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 5.00 volts.
 - b. Allow the voltage to decay over the current interval shown in the table.
 - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-13: 1.0T QR Series Ramping Profile

"Stage"	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	5.00	0 - 100	
2	4.40	100 - 200	
3	3.80	200 - 300	
4	3.00	300 - 400	
5	2.60	400 - 425	
6	2.00	425 - 450	
7	1.00	450 - 475	
8	0.30	475 - 497	
9	Slowly lower from 0.30 Volts to 0.10 Volts	497	
10	0.10	—	0.999
11	Set voltage as required to maintain constant field	Use "Recommended Site Parking Field" frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 10.9 - 11.3 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



WARNING

POTENTIAL MAGNET QUENCH!

1.0T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING.

ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 497 amps, slowly lower the Voltage Adjust Control from 0.3 V to 0.1 V Main Coil Voltage.
8. When the field strength reaches 0.999 Tesla, slowly adjust the Voltage Adjust Control to a 0.025 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.0T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 497 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.
10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).
15. Set the Heater 2 Shim Axial Switch to "O" (off).
16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

17. Remove the Ramp Lead Hold-Down Tool.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**! WARNING****POTENTIAL COLD BURN HAZARD**

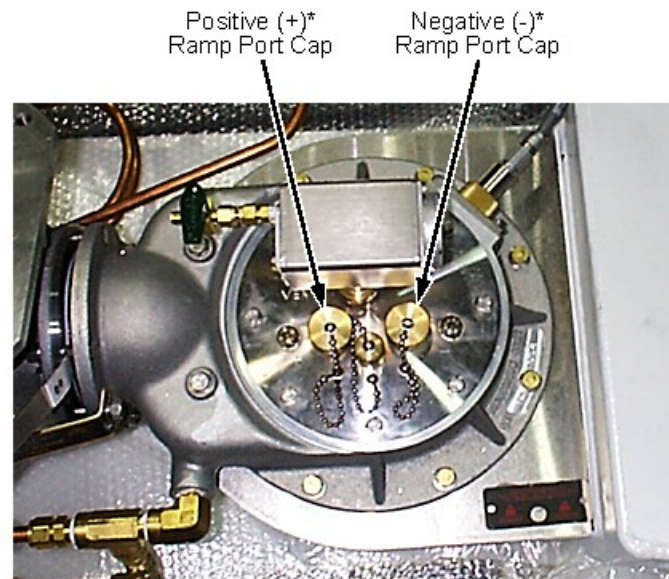
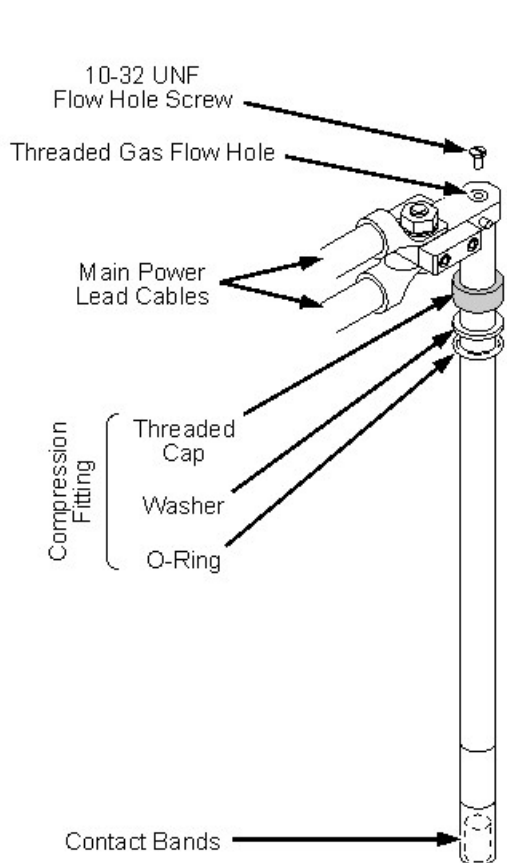
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:

- a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
- b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-85.

Illustration 3-85: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.

**NOTICE**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.

**NOTICE**

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.4.5 Finalization

No finalization steps.

7.4.5 1.0T Mobile Site Ramping

7.4.5.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.5.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.5.3 Preliminary Requirements

7.4.5.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.5.3.2 Consumables

No consumables required.

7.4.5.3.3 Replacement Parts

No replacement parts required.

7.4.5.3.4 Safety

No general safety information.

7.4.5.3.5 Required Conditions

No required conditions.

7.4.5.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: When both Main (Ramp) and Shim Power Supplies are used at a mobile site, the shim heater circuits of the two power supplies are connected in parallel. This puts a resistance load across the Main Coil Voltmeter of approximately 30 ohms. The Main Coil Voltmeter display will be approximately 90% of the actual voltage across the Main Coil. The ramp profile used in this subsection compensates for this condition.

Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:

- **Voltage Adjust Control:** zero (full CCW)
- **Current Adjust Control:** maximum (full CW)
- **Main Power Switches on the Main Ramp Power Supply:** "ON"
- **Main Power Switches on the Ramp Power Supply:** "ON"
- **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
- **Transverse 1 (T1) Shim Heater Switch on the Shim Power Supply:** "I" (ON)
- **Transverse 2 (T2) Shim Heater Switch on the Shim Power Supply:** "I" (ON)
- **Voltmeter Select Switch on the power supply:** "Main Coil" position

NOTE: The Axial Switch Heater current is supplied by the Main Service Ramp Supply. The Transverse 1 and Transverse 2 Switch Heaters currents are supplied by the Shim Supply. Do not use the Axial Heater on the Shim Supply as this will increase boil off.

2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.

3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).

4. Ramp the magnet in conformance with Table 3-14 as follows:

- a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 4.50 volts.

- b. Allow the voltage to decay over the current interval shown in the table.
- c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-14: 1.0T Mobile Site Ramping Profile

”Stage”	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	4.50	0 - 100	
2	4.14	100 - 200	
3	3.78	200 - 300	
4	3.42	300 - 400	
5	2.97	400 - 500	
6	2.25	500 - 550	
7	1.80	550 - 600	
8	1.35	600 - 625	
9	0.90	625 - 650	
10	0.68	650 - 675	
11	0.54	675 - 700	
12	0.45	700 - 720	
13	0.27	720 - 730	
14	Slowly lower from 0.27 Volts to 0.09 Volts	730	
15	0.09	—	0.999
16	Set voltage as required to maintain constant field	Use “Recommended Site Parking Field” frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 4.8 - 5.2 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



WARNING

POTENTIAL MAGNET QUENCH!

1.0T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 730 amps, slowly lower the Voltage Adjust Control from 0.27 V to 0.09 V Main Coil Voltage.
8. When the field strength reaches 0.999 Tesla, slowly adjust the Voltage Adjust Control to a 0.023 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.0T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 745 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).

16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

17. Remove the Ramp Lead Hold-Down Tool.



! WARNING

ELECTRICAL SHOCK HAZARD!

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.



! WARNING

ELECTRICAL SHOCK HAZARD!

DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.



! WARNING

POTENTIAL COLD BURN HAZARD

CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

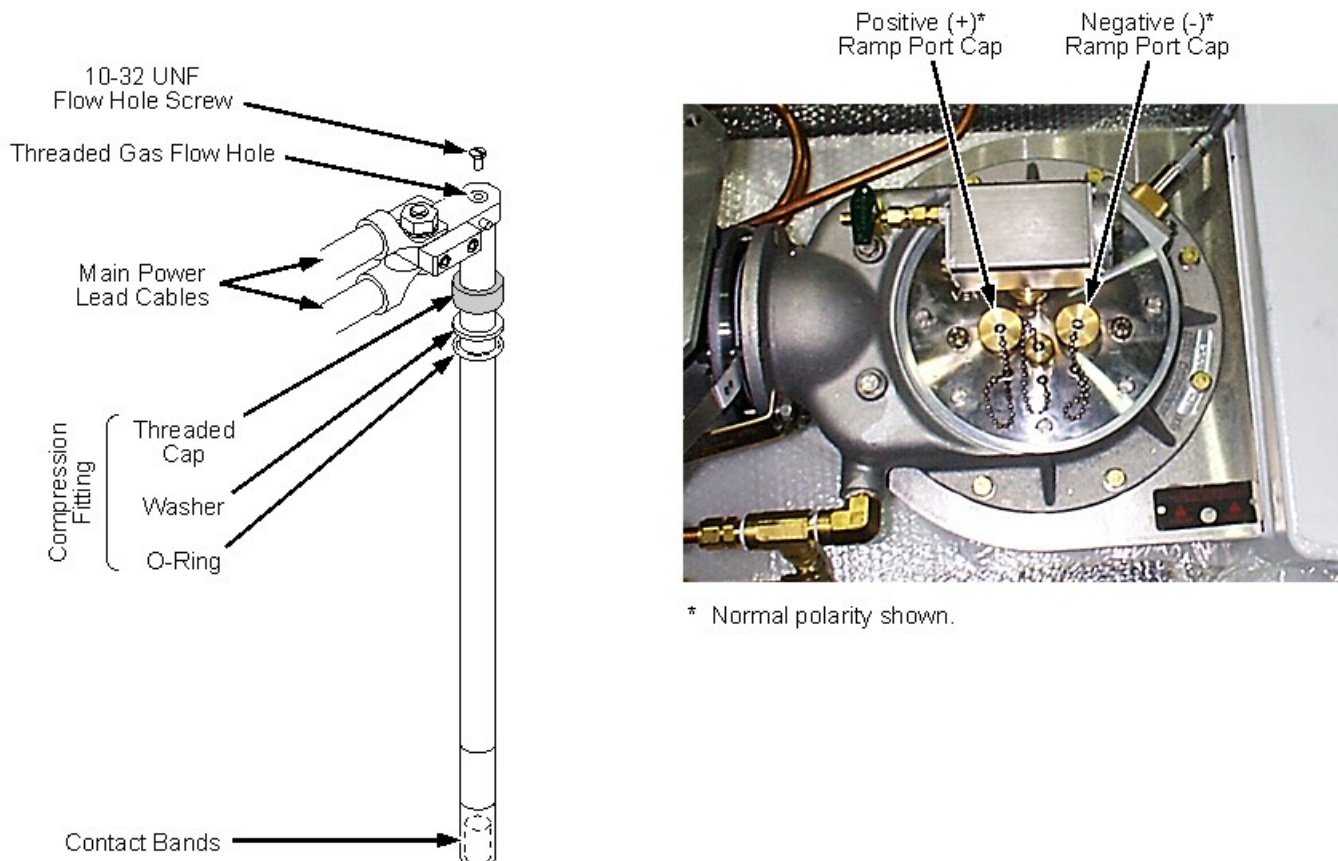


NOTICE

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:

- a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
- b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-86.

Illustration 3-86: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.

- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.



NOTICE

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.



NOTICE

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.5.5 Finalization

No finalization steps.

7.4.6 1.5T Mobile Site Ramping

7.4.6.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.6.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.6.3 Preliminary Requirements

7.4.6.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.6.3.2 Consumables

No consumables required.

7.4.6.3.3 Replacement Parts

No replacement parts required.

7.4.6.3.4 Safety

No general safety information.

7.4.6.3.5 Required Conditions

No required conditions.

7.4.6.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: When both Main (Ramp) and Shim Power Supplies are used at a mobile site, the shim heater circuits of the two power supplies are connected in parallel. This puts a resistance load across the Main Coil Voltmeter of approximately 30 ohms. The Main Coil Voltmeter display will be approximately 90% of the actual voltage across the Main Coil. The ramp profile used in this subsection compensates for this condition.

Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:

- **Voltage Adjust Control:** zero (full CCW)
- **Current Adjust Control:** maximum (full CW)
- **Main Power Switches on the Main Ramp Power Supply:** "ON"
- **Main Power Switches on the Ramp Power Supply:** "ON"
- **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
- **Transverse 1 (T1) Shim Heater Switch on the Shim Power Supply:** "I" (ON)
- **Transverse 2 (T2) Shim Heater Switch on the Shim Power Supply:** "I" (ON)
- **Voltmeter Select Switch on the power supply:** "Main Coil" position

NOTE: The Axial Switch Heater current is supplied by the Main Service Ramp Supply. The Transverse 1 and Transverse 2 Switch Heaters currents are supplied by the Shim Supply. Do not use the Axial Heater on the Shim Supply as this will increase boil off.

2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.

3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).

4. Ramp the magnet in conformance with Table 3-15 as follows:

- a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 5.00 volts.

- b. Allow the voltage to decay over the current interval shown in the table.
- c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-15: 1.5T Mobile Site Ramping Profile

”Stage”	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	5.40	0 - 100	
2	5.04	100 - 200	
3	4.68	200 - 300	
4	4.23	300 - 400	
5	3.87	400 - 500	
6	3.15	500 - 550	
7	2.70	550 - 600	
8	2.12	600 - 625	
9	1.80	625 - 650	
10	0.90	650 - 675	
11	0.68	675 - 700	
12	0.45	700 - 720	
13	0.27	720 - 730	
14	Slowly lower from 0.27 Volts to 0.09 Volts	730	
15	0.09	—	1.499
16	Set voltage as required to maintain constant field	Use “Recommended Site Parking Field” frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 10.9 - 11.3 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



⚠ WARNING

POTENTIAL MAGNET QUENCH!

1.5T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING.

ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 730 amps, slowly lower the Voltage Adjust Control from 0.27 V to 0.09 V Main Coil Voltage.
8. When the field strength reaches 1.499 Tesla, slowly adjust the Voltage Adjust Control to a 0.023 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.5T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 745 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).

16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

17. Remove the Ramp Lead Hold-Down Tool.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**! WARNING****POTENTIAL COLD BURN HAZARD**

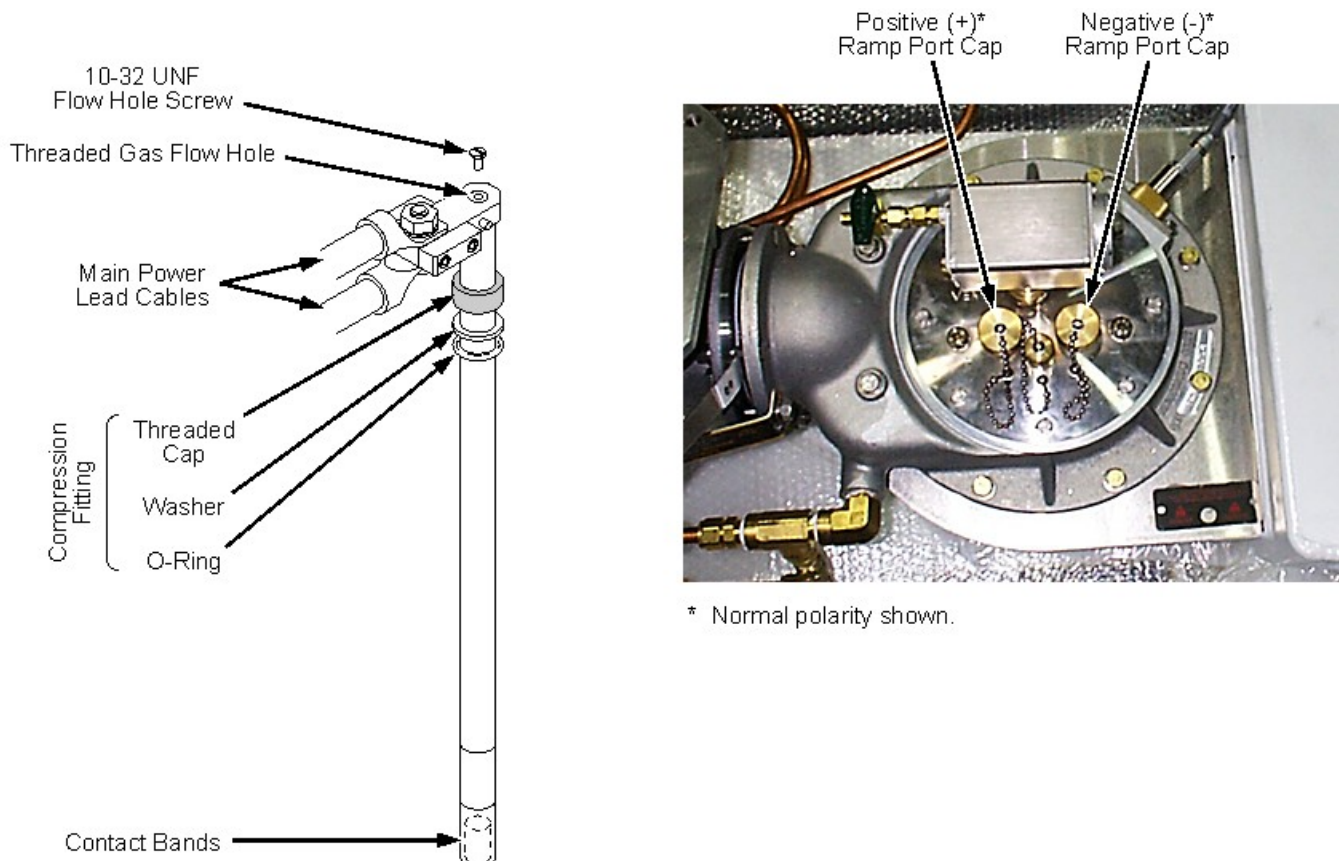
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:

- a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
- b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-87.

Illustration 3-87: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.

- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.



NOTICE

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.



NOTICE

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.6.5 Finalization

No finalization steps.

7.4.7 1.0T YQ Series Ramping

7.4.7.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.7.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.7.3 Preliminary Requirements

7.4.7.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.7.3.2 Consumables

No consumables required.

7.4.7.3.3 Replacement Parts

No replacement parts required.

7.4.7.3.4 Safety

No general safety information.

7.4.7.3.5 Required Conditions

No required conditions.

7.4.7.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:
 - **Voltage Adjust Control:** zero (full CCW)
 - **Current Adjust Control:** maximum (full CW)
 - **Main Power Switches on the Main Ramp Power Supply:** "ON"
 - **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
 - **Voltmeter Select Switch on the power supply:** "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).
4. Ramp the magnet in conformance with Table 3-16 as follows:
 - a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 2.50 volts.
 - b. Allow the voltage to decay over the current interval shown in the table.
 - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-16: 1.0T YQ Series Ramping Profile

"Stage"	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	2.50	0 - 100	
2	2.30	100 - 200	
3	2.10	200 - 300	
4	1.90	300 - 400	
5	1.65	400 - 500	
6	1.25	500 - 550	
7	1.00	550 - 600	
8	0.75	600 - 625	
9	0.50	625 - 650	
10	0.40	650 - 675	
11	0.30	675 - 720	
12	Slowly lower to 0.3 V	720	
13	0.30	—	0.999
14	Set voltage as required to maintain constant field	Use "Recommended Site Parking Field" frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 4.8 - 5.2 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



WARNING

POTENTIAL MAGNET QUENCH!

1.0T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING.

ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 720 amps, slowly lower the Voltage Adjust Control from 0.3 V Main Coil Voltage.
8. When the field strength reaches 0.999 Tesla, slowly adjust the Voltage Adjust Control to a 0.1 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.0T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 745 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.

13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).

16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

17. Remove the Ramp Lead Hold-Down Tool.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**! WARNING****POTENTIAL COLD BURN HAZARD**

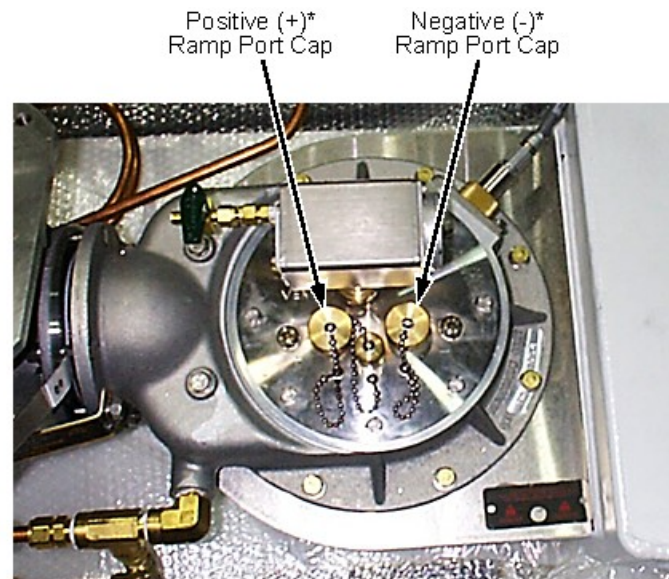
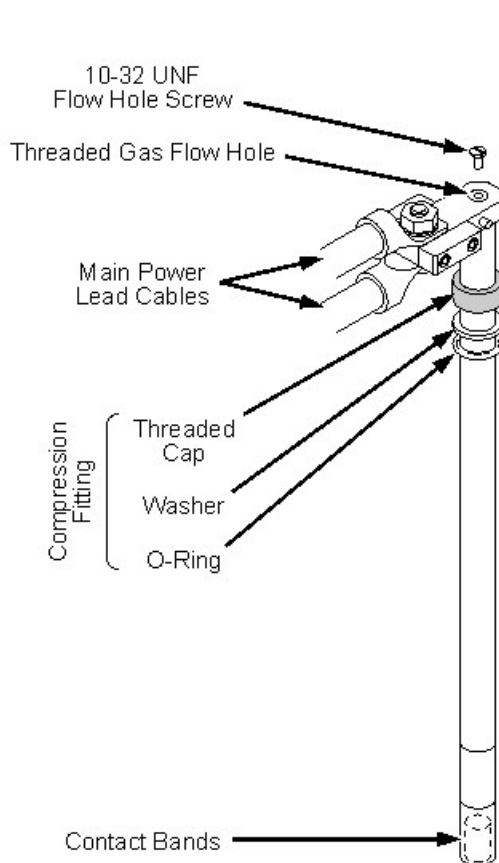
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:

- a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
- b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-88.

Illustration 3-88: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.

**NOTICE**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

NOTE: If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply **OR**, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters. Both the Axial and Transverse Heaters are available on the Shim Power Supply.

**NOTICE**

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.7.5 Finalization

No finalization steps.

7.4.8 1.5T YR Series Ramping

7.4.8.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	Not Applicable	Not Applicable

7.4.8.2 Overview

See Section 7.2, Ramping Preparations, for “Personnel Requirements” and “Preliminary Requirements” for the entire ramping procedure.

7.4.8.3 Preliminary Requirements

7.4.8.3.1 Tools and Test Equipment

No tools or test equipment required.

7.4.8.3.2 Consumables

No consumables required.

7.4.8.3.3 Replacement Parts

No replacement parts required.

7.4.8.3.4 Safety

No general safety information.

7.4.8.3.5 Required Conditions

No required conditions.

7.4.8.4 Procedure



WARNING

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH AND THE COILS BECOME DAMAGED IF THE AXIAL SWITCH HEATERS ARE NOT “ON” DURING RAMPING.

MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE “ON” DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. (GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.)



CAUTION

Potential Magnet Quench

A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the vent system.

If a quench occurs during ramping, immediately: a) turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW), b) turn off the Power Supply, c) leave the room and d) turn off the Cryocooler Compressor outside the magnet room. Do not reenter the magnet room until the quench is over.

NOTE: Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure the power supply controls are set as follows:
 - **Voltage Adjust Control:** zero (full CCW)
 - **Current Adjust Control:** maximum (full CW)
 - **Main Power Switches on the Main Ramp Power Supply:** "ON"
 - **Heater 2 Shim Axial Switch on the Main Power Supply:** "I" (ON)
 - **Voltmeter Select Switch on the power supply:** "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength. The probe will lock on when magnet current rises to approximately 515 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 - 29.803550 MHz).
4. Ramp the magnet in conformance with Table 3-17 as follows:
 - a. Slowly turn the power supply's Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 3.00 volts.
 - b. Allow the voltage to decay over the current interval shown in the table.
 - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.

Table 3-17: 1.5T YR Series Ramping Profile

"Stage"	Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range	Field Reached (Tesla)
1	3.00	0 - 100	
2	2.80	100 - 200	
3	2.60	200 - 300	
4	2.35	300 - 400	
5	2.15	400 - 500	
6	1.75	500 - 550	
7	1.50	550 - 600	
8	1.20	600 - 625	
9	1.00	625 - 650	
10	0.50	650 - 675	
11	0.35	675 - 720	
12	Slowly lower from 0.35 Volts to 0.30 Volts	730	
13	0.30	—	1.499
14	Set voltage as required to maintain constant field	Use "Recommended Site Parking Field" frequency stated in ATR	

5. When the Main Coil (Ramp) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L_{\text{Inductance}} = 10 \times \text{Voltage} / \text{Current Change}$$

NOTE: This method will give inaccurate values of inductance when the current is less than 200 amps.

6. If inductance (L) is outside the range of 10.9 - 11.3 henries, discontinue ramping, measure main coil resistance in conformance with Chapter 4, Section 5, Magnet Electrical Checks, and contact your regional MAC Team representative.



⚠ WARNING

POTENTIAL MAGNET QUENCH!

1.5T LCC MAGNETS CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING.

ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEPS.

7. When the magnet current reaches 730 amps, slowly lower the Voltage Adjust Control from 0.3 V to 0.1 V Main Coil Voltage.
8. When the field strength reaches 1.499 Tesla, slowly adjust the Voltage Adjust Control to a 0.1 V Main Coil Voltage.

NOTE: The magnet's ATR gives the optimum magnet on-site parking field beside "Recommended Site Parking Field." The parking field is generally 25 Gauss above the final field of 1.5T to compensate for correction coil currents.

9. Check the Teslameter reading and **slowly** adjust the Voltage Adjust Controls as required to bring the magnetic field to frequency shown on the ATR as the "Recommended Site Parking Field." The total current will be approximately 742 amps. Allow the field to stabilize. The last two digits on the Teslameter should be the only digits changing.

10. Maintain field at Step 9's setting for 5 minutes before proceeding to Step 11.

NOTE: Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in "Main Coil" position. When the switch goes 'persistent' the voltage across the magnet terminals will stabilize at 0.00.

11. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'



WARNING

POTENTIAL MAGNET QUENCH!

INCORRECT MAIN COIL CONNECTION POLARITY AND RAMPING CURRENT SETTINGS AT A LATER DATE CAN RESULT IN A MAGNET QUENCH WHEN THE MAIN POWER SUPPLY SWITCH IS TURNED ON.

WHEN CHANGING MAGNETIC FIELD IN THE FUTURE, THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS. UPON COMPLETION OF EACH RAMP RECORD MAIN COIL CONNECTION POLARITIES AND FINAL RAMPING CURRENTS IN "MAGNET RAMPING AND PARKING CURRENT LOG" IN THE DATA SHEETS CHAPTER.

12. Record current, frequency and lead extension voltage values at which the switch went 'persistent' in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.

13. When the switch goes 'persistent,' **slowly** turn the power supply's Voltage Adjust Control to zero (full CCW) over a two minute period.

NOTE: Check that the Teslameter does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Adjust Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

14. Gradually turn the Current Adjust Control to zero (over a one minute period).

15. Set the Heater 2 Shim Axial Switch to "O" (off).

16. Set the Main Power and Power On switches on the Main Power Supply to "OFF" and disconnect the Input Power Cable.

17. Remove the Ramp Lead Hold-Down Tool.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.

**! WARNING****ELECTRICAL SHOCK HAZARD!**

DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**! WARNING****POTENTIAL COLD BURN HAZARD**

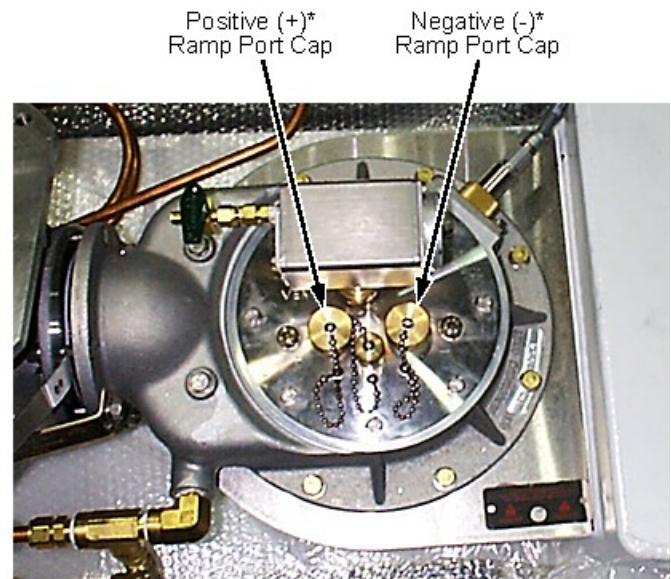
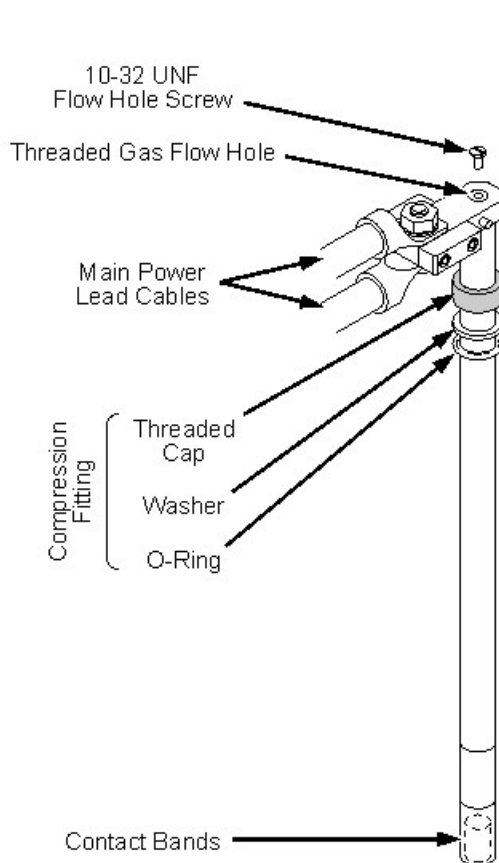
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**NOTICE**

Step 18 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

18. After parking the magnet and disconnecting the main power supply, plug and **remove one Main Extension at a time** in the following sequence:

- a. Open Valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the magnet pressure gauge or Magnet Monitor. Close V2.
- b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed first. See Illustration 3-89.

Illustration 3-89: Main Lead Extension and Ramp Ports

* Normal polarity shown.



⚠ WARNING

POTENTIAL MAGNET QUENCH!
PLUGGING THE FLOW HOLE OF AN ENGAGED MAIN LEAD EXTENSIONS WILL CAUSE THE MAGNET TO QUENCH.
DO NOT INSERT FLOW HOLE SCREWS INTO MAIN LEAD EXTENSIONS UNTIL AFTER EXTENSIONS ARE REMOVED.

- c. Unscrew the ramp lead port compression fitting and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
- e. Unscrew the ramp lead port compression fitting and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
- f. Install the flow hole screws in the flow hole of both Main Lead Extensions **after** both leads have been removed.
- g. Check the Main Leads and fill port cap for leaks.

**NOTICE**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve V2.

19. Plug in the heater on the Magnet Monitor with a setpoint of 4.0 psig. Allow the field to settle.

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**NOTICE**

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with Section 8.2, Preparations for Shimming, Magnet Stability Check, before centering the Teslameter Probe and shimming the magnet.

20. Proceed to Section 8.1, Shimming.

7.4.8.5 Finalization

No finalization steps.